Physical-Only Proprioceptive Recognition of Fifteen Classes with a LEAP Hand

G. Sai Pranav, Harish J. Palanthandalam-Madapusi

Abstract

Robotic hands commonly use vision or dedicated tactile sensors to recognize objects. Using 15 household objects, this study tests whether a physical LEAP Hand can distinguish between objects using only the joint angles reached after controlled grasps. Motor-current thresholds stop an entire finger when resistance is detected. Three sequential grips on the object provide 48 final angles per trial to a 24-unit hidden-layer classifier. The dataset contains five physical trials for each of 15 labels, comprising 14 objects and an empty-hand condition. In three-fold within-dataset validation, the neural network classified 70 of 75 held-out trials correctly. Its decision rule accepted 67 trials as known classes, of which 65 were correct. A nearest-trial baseline obtained 69 of 75 correct in leave-one-trial-out evaluation. The class set was selected after preliminary analysis, so these results are internal, post-selection estimates. They do not measure generalization to a new session, object instance, or unseen class.

Keywords: proprioception, active perception, LEAP Hand, household object recognition, neural network, physical grasping

1. Introduction

A robot can learn about an object by interacting with it. For a dexterous hand, the joint positions reached while closing around an object depend on the surfaces that interrupt finger motion. These positions may therefore contain information about the object even without an external camera or dedicated tactile skin. This use of the hand's internal state is known as proprioceptive perception. The LEAP Hand is a dexterous, anthropomorphic hand developed for robot learning and manipulation Shaw et al. (2023) and references therein.

Previous work has shown that joint information can support object recognition. Vásquez and Perdereau (2017) developed proprioceptive shape signatures from robotic-hand joint angles and studied shape identification across object sizes and poses. Sintov and Meir (2023) used grasp information for kinesthetic object recognition without dedicated tactile sensors. More recently, Kutsuzawa et al. (2024) investigated learning-based shape and stiffness estimation from joint angles and velocities in a multi-fingered hand. These studies motivate a physical-hand experiment using a simple observation: the final joint angles of a repeatable grip sequence.

The main research question is whether final angles from the physical LEAP Hand can distinguish different objects, including an empty-hand condition. A second question is whether combining repeated grip sequences might improve decisions that are uncertain after one sequence; the current dataset evaluates individual sequences only. The model predicts a class label, rather than object size. Each object label represents the recorded physical exemplar; recognition of another instance of the same kind is a separate generalization question.

The implementation provides repeatable grips, a resumable physical-data collection procedure, a neural classifier, a nearest-trial comparison method, and an interface for running the hand and viewing the results. Both models use only physical grasp measurements for training and recognition.

2. Experimental Setup and Methodology

2.1. Physical hand and controlled grips

The protocol uses a four-fingered LEAP Hand with 16 actuated joints, four per finger (Shaw et al., 2023). The fingers are the index, middle, ring, and thumb. All angles are expressed in degrees. Each object is placed near a consistent point relative to the palm, and every object receives the same programmed grip sequence. The saved trial files do not record measured placement offsets or object orientation. Placement is therefore a condition of the collection procedure rather than a property inferred by the model.

For each configured grip, the hand moves to that grip's starting posture. Its 16 joints then advance toward specified targets using the per-joint step sizes in the experiment settings. During closure, motor current served as an indicator of resistance. If a moving joint's signed current exceeds its configured threshold, all four joints of that finger stop at their measured positions. The remaining fingers continue until resistance or an angle target

stops them. A grip ends when all four fingers have stopped and is aborted after at most 250 iterations. The hand moves to the next grip's starting posture before continuing the sequence.

After each grip, the program reads its final 16 joint angles, retrying for up to 10 seconds if position reporting is temporarily unavailable. The final vectors are concatenated in grip order. The hand runs three distinct grips, yielding 48 input angles per physical trial.

During collection, the final angles and time-series log are saved before the hand returns to its starting posture. Motor currents control the movement but are not classifier inputs. No camera images or tactile arrays are used as recognition features.

2.2. Physical dataset

Each training record consists of an enrolled class label and one concatenated final-angle vector,

$$\mathbf{q}_i = \left[\mathbf{q}_i^{(1)}, \ldots, \mathbf{q}_i^{(G)}\right] \in \mathbb{R}^{16G},$$

where G is the number of configured grips and each $\mathbf{q}_i^{(g)}$ contains the 16 final joint angles from grip g. The data-collection program stores every completed trial with its final angles and time-series log. It also saves a collection plan before recording begins, so interrupted sessions can resume without discarding earlier trials. Additional trials and new classes can be added later. A mistaken trial may be retaken; the previous record is archived after a complete replacement has been checked.

The dataset has 15 labels: water bottle, Rubik's cube, ball, bowl, mug, basic cylinder, power bank, spectacles case, mouse, stapler, pencil box, Pyraminx, tape measure, spice bottle, and no object. Fourteen labels identify physical items and one identifies an empty-hand grasp. Five recorded trials per label give 75 physical trials. The trial files contain the class name, final angles, and time-series motor log; they do not contain measured object dimensions or placement coordinates. The analyzed class set was selected after preliminary analysis of recorded grasps rather than fixed before data collection.

2.3. Neural network and training

The neural network receives only the $16G$ final joint angles. Each angle is normalized by the start and target angles of its grip and joint:

$$x_{g,j} = \frac{q_{g,j} - a_{g,j}}{b_{g,j} - a_{g,j}},$$

where $a_{g,j}$ and $b_{g,j}$ are the start and target angles for joint j in grip g. A joint with equal start and target values is assigned a normalized value of zero.

The classifier has $16G$ input units, one fully connected 24-unit hidden layer with a hyperbolic tangent activation, and one output for each enrolled class. A softmax operation converts the outputs to class probabilities. For C classes, the model has $384G + 24 + 25C$ trainable parameters. With three grips and 15 classes, it has 1,551 parameters. The model does not predict object size.

Training uses full-batch Adam optimization with a learning rate of 0.01, weight decay of 0.001, and a limit of 500 epochs. Each real trial is retained and supplemented with 20 noisy copies during training. Independent zero-mean Gaussian noise with a standard deviation of 0.5° is added to each angle. A second 0.5° Gaussian perturbation is shared by the four joints of each finger within each grip. These artificial copies approximate small measurement and grasp variation; they are not counted as additional physical trials. Samples are weighted inversely to the number of real trials in their class to reduce the effect of unequal class sizes.

2.4. Validation and recognition decisions

The implemented evaluation uses up to three folds, with real trials from every class assigned to different folds before noise is added. Noise augmentation is performed only on each training fold. The validation fold therefore contains measurements from the physical hand, not perturbed copies of training samples. The best epoch in each fold is selected by validation cross-entropy, with training stopped after 80 epochs without improvement. The final model is then trained on all real trials for the median number of selected epochs.

The class probabilities are adjusted with a temperature selected from the held-out folds. We also compute the distance from a new normalized angle vector to the nearest saved physical training grasp. A prediction is reported as *unknown* when this distance exceeds 1.5 times the 95th percentile

of the validation distances. Otherwise, it is reported as *uncertain* when the largest class probability is below 0.75 or the difference between the two largest probabilities is below 0.20. A known class is returned only when these checks pass. These thresholds are calibrated from known-class validation trials. Their ability to reject genuinely unseen objects requires a separate evaluation.

If a neural prediction remains uncertain, the program can request another trial of the same object, up to three trials. Each trial runs the configured grip sequence. The class probabilities are averaged across trials, while the nearest-reference distances are combined using their median. The resulting decision is saved with each trial. A separate nearest-trial method assigns the class of the saved physical grasp with the smallest Euclidean distance in degrees. That method supplies a comparison baseline; it does not produce probabilities or an unknown-object decision.

2.5. Software implementation

The grip and physical-only training programs are part of Ex-ARM (Pranav, 2026). The terminal interface can collect, resume, extend, and retake physical trials, train either model, and run the hand for recognition. The presentation window evaluates both trained models from the same recognition grasp. It shows both decisions, the neural probabilities or nearest-trial distances, and a visualization driven by the measured hand pose. The classifier itself uses only recorded physical angles. The repository retains the trial CSV files and model reports, including the neural confusion matrices and nearest-trial evaluation results.

3. Results and Discussion

3.1. Within-dataset validation

Each trial was predicted once by a three-fold held-out neural model before the final model was trained on all recordings. Raw accuracy counts the largest softmax probability even when the decision rule would instead return *uncertain* or *unknown*. The accepted count includes only decisions assigned an enrolled class. The results are summarized in Table 1.

The neural model correctly classified 70 of its 75 held-out trials before rejection. It accepted 67 as known classes, of which 65 were correct; the other eight trials were uncertain. Table 2 gives the raw result for every label. Ten labels had five correct predictions in five trials. The five remaining labels

Table 1: Internal validation of the 15-class physical dataset. Accepted correct is conditioned on a known-class decision. The nearest-trial method uses leave-one-trial-out evaluation rather than the neural network's three folds.

Measure	15 classes
Recorded physical trials	75
Neural raw correct	70/75 (93.3%)
Accepted known-class decisions	67/75 (89.3%)
Accepted correct	65/67 (97.0%)
Uncertain decisions	8
Unknown decisions	0
Nearest-trial correct	69/75 (92.0%)

each had one error. Because all labels have five trials, macro-averaged recall equals raw accuracy at 93.3%.

The nearest-trial comparison classified 69 of 75 trials correctly. This baseline removes each tested trial from its reference set, while the neural result holds out approximately one third of the real trials per fold and uses those folds for epoch selection. The scores therefore describe different internal evaluations; their small numerical difference is not a controlled comparison on a separate test set.

3.2. Interpretation and limitations

The internal results support the presence of class information in the final joint configuration. They do not establish performance on another recording day, a different exemplar of a named object, or an object absent from training. No separate novel-object rejection experiment is recorded in these reports.

The 15-class set was selected after preliminary analysis of the recorded grasps. Its validation score is therefore a within-dataset, post-selection estimate rather than an independent test of a prospectively fixed class set. Likewise, the probability temperature and distance threshold were chosen from the held-out folds. Accepted accuracy on those folds may be optimistic for new data. Only five trials are available for each label; a single error changes that label's measured recall by 20 percentage points.

The final joint configuration is not a unique description of an object's shape. Items of different appearance may produce similar angles at the same palm location. Gaussian angle perturbations do not create independent physical grasps and cannot reproduce every change in placement, friction, compliance, or motor-current response. The software permits up to three grip

Table 2: Raw held-out neural predictions in the 15-class study. The table reports the highest-probability label before the uncertainty rule is applied.

True label	Correct	Incorrect prediction
Water bottle	4/5	Mug
Rubik's cube	5/5	—
Ball	5/5	—
Bowl	5/5	—
Mug	4/5	Ball
Basic cylinder	4/5	Rubik's cube
Power bank	4/5	Spice bottle
Spectacles case	5/5	—
Mouse	4/5	Bowl
Stapler	5/5	—
Pencil box	5/5	—
Pyraminx	5/5	—
Tape measure	5/5	—
Spice bottle	5/5	—
No object	5/5	—

sequences for an uncertain recognition and averages neural probabilities, but the saved validation reports evaluate individual sequences; they do not measure the benefit of repetition. A future independent test should fix the class set, model, and thresholds before collecting labeled grasps in later sessions, include new object instances, and reserve some entire object classes to measure unknown rejection.

4. Conclusion

This work demonstrates physical-only proprioceptive recognition with a LEAP Hand using three sequential grips and a 48-angle neural input. In internal three-fold validation, the 15-class model classified 70/75 trials correctly and accepted 67/75 as known-class decisions. Of the accepted decisions, 65 were correct. The nearest-trial baseline provided a separate leave-one-trial-out comparison. These measurements show that final grasp angles distinguish many enrolled items under the recorded conditions. Independent sessions, new physical instances, unseen objects, and the effect of repeat grasps remain to be tested.

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